

Name: Student ID:

ECS401: Unit 10 Lab Progress Test

These exercises are to help you judge your progress. It is really important that you make sure you fully understand everything we have covered so far.

Consider the following `bubblesort` method where the method `swap` swaps values in the array at the given positions when answering these questions:

```
public static void bubblesort(int[] array)
{
    int n = array.length;
    for(int pass=0; pass <= n-2; pass++)
    {
        for(int pos= 0; pos <= n-2-pass; pos++)
        {
            if (array [pos] > array [pos + 1])
                swap (array, pos, pos + 1);
        }
    }
}
```

1. If given an array of length 450, how many passes will `bubblesort` need to guarantee to sort the data in to order?

Answer:

2. How many comparisons will be done on the very **first** pass if this `bubblesort` algorithm is given an array of length 450?

Answer:

3. How many comparisons will be done on the very **last** pass if this `bubblesort` algorithm is given an array of length 450?

Answer:

4. Assuming the above algorithm, what is the sequence of states the array passes through during the first pass if given the array $a = \{4, 3, 1, 6, 9, 5\}$?

a[0]	a[1]	a[2]	a[3]	a[4]	a[5]
4	3	1	6	9	5

5. What is the state of the array AFTER each pass of bubblesort is completed if given the initial array $a = \{6, 9, 3, 5, 4, 1\}$?

PASS	a[0]	a[1]	a[2]	a[3]	a[4]	a[5]
START	6	9	3	5	4	1
0						
1						
2						
3						
4						

6. What would the line

```
if (array [pos] > array [pos + 1])
```

be changed to if the array was to be sorted in to descending order rather than ascending order?

Answer:

7. Give a version of the algorithm to sort String arrays in to alphabetical order. You may assume you have an appropriate swap method.



Answer:

8. Give a version of the algorithm to sort an array of Student records (of type Student) in ascending order based on the mark field. You may assume you have an appropriate swap method.



Answer: